

Intelligent Systems - Machine Learning (IS-ML)

MICS2-62

Take-Home Assignment 1 (THA1)

Grading scheme: 20 points (0-20)

Weight for final grade: 10%

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Release date: October 5th, 2026.

Deadline: November 2nd, 2026, 23:59

Preamble

Please submit your solutions (the developed code and a report in PDF format) through Moodle. State clearly in the report the group number and the group members.

The solutions for the Take-Home Assignment 1 can be submitted just by a group. Individual submissions are not allowed. Each group shall have three members.

Before starting, please check the Large Language Models (LLM) usage policy at <https://www.uni.lu/wp-content/uploads/sites/9/2026/02/25170716/2026-Guidelines-AI-I2TL-16022026.pdf>. Please be aware that none of the exercises here require the use of an LLM.

A. Supervised Learning (SL) - Regression (7.5 points)

Problem

In this exercise (A) you are requested to perform linear and polynomial regression on the data collected from a Combined Cycle Power Plant. The data can be found in Moodle in the file "Data Take Home Assignment 1 Exercise A.xlsx". It contains two columns. The first column represents the *Temperature* (the input data - X) on the Celsius scale and the second column represents the *Net hourly electrical energy output* (the output data - Y) in MW. Each row represents a data point.

Requirements

- Please solve each of the following subproblems and give clear explanations for your solutions.
- For this exercise (A), you are requested to develop the code for Least Squares, Linear Regression and Polynomial Regression optimised with Gradient Descent from scratch. That is: do not use any libraries such as Scikit-Learn to implement these machine learning models for regression, the equations that you will report in the pdf report to calculate the derivative of the loss functions, the gradients, the update rules, and so on, shall be clearly identifiable in the developed code.

A.I. Linear regression (4.5 points)

1. (0.5 points) *Data acquisition*. Each group has to perform linear regression on 20 data points (not on all data points from the file). Assuming that your group number is n then the data points

corresponding to your group starts with row number $(n - 1) * 20 + 2$ and ends with the row number $n * 20 + 1$ (e.g. Group 1 has the data from row 2 until row 21, Group 2 has the data from row 22 until row 41, and so on). Present your data in a table in the report.

2. (0.4 points) *Data transformation*. Perform a transformation of your acquired data. Please feel free to use any type of transformation you prefer (e.g. Min-Max normalization, Z-score standardization). Report the formula that you used for the transformation and the transformed data in the report. Justify your choice.
3. (0.8 points) *Least Squares*. Perform linear regression on your data using least squares (closed form solution). Print the obtained regression parameters. Plot the regression line and the data points.
4. (0.8 points) *Linear Regression with Gradient Descent - cost function*. Choose a suitable cost function to perform linear regression with gradient descent. Motivate your choice. Perform the partial derivatives of the cost function with respect to each of the regression parameters. Report also the update rules.
5. (0.4 points) *Linear Regression with Gradient Descent - first iteration*. Start from a random choice of the regression parameters (print the choice in the report) and perform the first iteration of gradient descent on your data. Report the chosen learning rate and the computed values. Plot the regression line and the data points.
6. (0.4 points) *Linear Regression with Gradient Descent - second iteration*. Perform the second iteration of gradient descent. Report the computed values. Plot the regression line and the data points.
7. (0.4 points) *Linear Regression with Gradient Descent - last iteration*. Continue performing gradient descent for an arbitrary number of iterations until the regression line fits well the data. Plot the values of the cost function for all iterations. Report the regression parameters and plot the regression line that you obtained after the last iteration.
8. (0.8 points) *Discussion*. Compare the performance of Least Squares and Linear Regression optimised with Gradient Descent. Discuss the similarities and the differences between them.

A.II. Polynomial regression (3 points)

- *Discussion*. Using the data from subproblem numbers A.I.1 or A.I.2 (this is your choice, but please justify it) please derive, implement, and perform polynomial regression using the following details:
- *Regression model*: $h_{\theta}(\mathbf{x}) = \theta_2 x^2 + \theta_1 x + \theta_0$, where $\theta_2, \theta_1, \theta_0$, are the model parameters
- *Cost function*: $J(\theta) = \frac{1}{4n} \sum_{i=1}^n (y^{(i)} - h_{\theta}(\mathbf{x}^{(i)}))^4$, where i iterates over all data points n , $\mathbf{x}^{(i)}$ represents the input of a data point (i) , $\mathbf{y}^{(i)}$ represents the true output of a data point (i)
- *Optimization method*: Gradient descent

The report shall contain:

1. (0.4 points) Partial derivatives of the cost function with respect to each of the model parameters.
2. (0.4 points) The update rule for each parameter
3. (1.5 points) Plots to reflect how during the gradient descent iterations the model fits better and better the data. Stop the iterations when the model fits well the data. Please report also the initial values of the model parameters and the learning rate.
4. (0.7 points) A discussion with the differences and similarities between linear and polynomial regression.

B. Supervised Learning (SL) - Classification (5.5 points)

Problem

In the exercise B.I. you are requested to create two artificially generated datasets (as detailed below) and to use the scikit-learn library to visualize the decision boundaries of k-NN, Naive Bayes, and Support Vector Machine (SVM). Even if it was not discussed during the lectures, please feel free to add also a Decision Tree and an Artificial Neural Network classifier.

After it, in exercise B.II. you are asked to study about a concept called bias-variance tradeoff and relate it to k-NN.

Requirements

- Please solve each of the following subproblems and give clear explanations for your solutions.
- For this exercise (B), you do not have to implement the machine learning models from scratch. Some hints: try to use the scikit-learn library; this tutorial https://scikit-learn.org/stable/auto_examples/classification/plot_classifier_comparison.html can be used as a guideline to solve this exercise.
- It is common to have multiple implementations of a method in scikit-learn (e.g. MultinomialNB, GaussianNB, ComplementNB, etc.). In such cases, choose a single implementation for this method and justify your reasoning about it. For Support Vector Machine in specific, we recommend using the C-Support Vector Classification (SVC) implementation¹ because of exercise B.I.5.

B.I. Classification models comparison (4 points)

1. (0.4 points) Dataset 1 creation. Create a dataset for a binary classification problem. It shall have as input two continuous real number attributes, and 40 data points belonging to one class and 40 data points belonging to the other class. Discuss your approach and report the data points obtained in the report.
2. (0.4 points) Dataset 2 creation (Dataset 1 + outliers). Add randomly to Dataset 1, four outliers for each class. Discuss your approach and report and visualize the data points, including the outliers in the report.
3. (0.4 points) Split Dataset 1 and Dataset 2 in training and testing data. Justify your choices for splitting the data and report the split data.
4. (0.8 points) Train on Dataset 1 and Dataset 2, the following classification models: k-NN, Naive Bayes and Support Vector Machine. Report the classification accuracy obtained on the training and on the testing data. Report also the confusion matrix. Discuss the obtained results.
5. (0.4 points) Explore different values for the regularization parameter and different kernels for the Support Vector Machine classifier. Do they affect the accuracy? Discuss the observed results.
6. (0.8 points) Taking inspiration from the tutorial mentioned above, visualize the decision boundaries of the various classification models used, including at least two different kernels for SVC.
7. (0.8 points) Discuss the similarities and the dissimilarities between the various classification models. Include in the discussion also their decision boundaries.

¹<https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>

B.II. Bias-Variance tradeoff (1.5 points)

Using the code that was built in exercise B.I. we will now explore a concept called “bias-variance trade-off” using the k -NN classifier.

Familiarize yourself with this concept and follow the instructions below. A good starting point for this concept can be found on section 3.2 of *Pattern Recognition and Machine Learning* by C. M. Bishop or in section 2.2.2 of *An Introduction to Statistical Learning with Applications in R* by G. James, D. Witten, T. Hastie, and R. Tibshirani.

1. (0.5 points) Train 5 different k -NN models using $k \in \{1, 3, 5, 20, 50\}$ on Dataset 1. Plot the decision boundaries for all 5 models (using subplots, for example). Repeat this visualization process for Dataset 2. Discuss how increasing k impacts the decision boundaries for both datasets.
2. (1 point) Using the results observed in B.II.1, explain the bias-variance tradeoff. Identify which values of k correspond to high variance / low bias and which correspond to high bias / low variance. Feel free to use the plots of the previous exercise to support your argument. Lastly, relate overfitting and underfitting to the value of k in this experiment.

C. Unsupervised Learning (UL) - free choice study (4 points)

Perform a study at your choice on one typical unsupervised learning task. Examples of unsupervised learning tasks are clustering, dimensionality reduction, unsupervised feature selection, unsupervised feature extraction, density estimation, and so on. Please feel free to choose the task that you will work on and the corresponding dataset(s) based on your personal research interests. Motivate your choice.

For the chosen task please address the following aspects (C.I) in your report and code. While making your choices, and addressing these aspects, please feel free to be creative and think out of the box. Always motivate your choice. The report shall contain for this exercise (C) good qualitative discussions and references to the literature for the machine learning models used, the various design choices, and statements (except for the ones derived from your own results) that you made. Please acknowledge also the code repositories that you used.

C.I. Instructions

1. (0.4 points) Choose the UL task and suitable evaluation metrics for it. Present and motivate your choices in the report.
2. (0.4 points) Choose a suitable dataset. Discuss (and visualize - if possible) in the report the data, the training/validation/testing split, and any other operations performed on your original data. Motivate your choice.
3. (0.8 points) Choose two machine learning models (or more - up to you) suitable to address the chosen UL task. Motivate your choice in the report.
4. (0.8 points) Discuss and present in the report how the chosen machine learning models work. Present their corresponding optimization methods and discuss also their implementations in the report (please feel free to use implementations from open-source libraries or to implement the code yourself - up to you).
5. (1.2 points) Train the machine learning models on your data, present the results, and discuss them in the report. If there are any interesting findings or observations please allocate sufficient space to discuss them qualitatively.
6. (0.4 points) Present in the report the conclusion of your work on this exercise (C) and discuss the work's limitations and possible future improvements.

Hints:

- Please feel free to use just the ML models and UL tasks discussed during our unsupervised learning lecture or to pick completely different ML models and UL tasks from the literature. Both options are fine and the choice is up to you.
- This exercise (C) is very flexible. Try to have a systematic approach. Choose small datasets and machine learning models which require low computational resources in order to be able to run them easily on a typical laptop.
- To avoid writing too much, it would be probably sufficient to allocate in your report 3-6 pages for this exercise (C), except the list of references which can be as long as it needed.

D. Model evaluation and Validation strategies (3 points)

In this problem you will explore different validation techniques and evaluate how different resampling methods affect model selection.

1. (0.5 points) Split Dataset 1 from exercise B.I.1 into a training dataset containing 80% of the data and a test set with the remaining 20%. Fit three SVC models, each one with a different value for the regularization factor $C \in \{0.01, 1, 100\}$.

Assess the accuracy of each model on the training set and report it on a table.

2. (1 point) Using the same training dataset and regularization factors, train these models using a 5-fold cross validation². Report on a table the average training accuracy of the 5 folds as well as the standard deviation for each one of the three models.

Compare the performance metrics between the single split and cross-validation and discuss which one provides a more reliable estimate and why.

3. (1.5 points) Finally, select one the model with the best value of C (among the values explored) and re-evaluate it once on the held-out test set from Part 1. Explain why this model was chosen and report its test accuracy. Compare this observed test accuracy with the train accuracy from exercise D.1 and with the train accuracy from D.2.

Success!

²This method is commonly known as k-fold cross validation. An interesting resource to understand it can be found at https://scikit-learn.org/stable/modules/cross_validation.html up to section 3.1.1.1.